

Speed, Distance and Time — Paper A

MEDIUM • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every answer must carry a unit. An answer with no unit does not score the final mark.
- The mark for each question is shown in square brackets on the right.
- Reminder: $\text{speed} = \text{distance} \div \text{time}$, and to convert m/s to km/h multiply by 3.6.

SECTION 1 – THE FORMULA

- 1.** Write down the formula for speed, and state the SI unit of speed. [2]

- 2.** Calculate the speed in each case.

- (a)** 100 m travelled in 20 s. [1]

- (b)** 240 km travelled in 3 hours. [1]

- (c)** 45 m travelled in 6 s. [2]

- 3.** Calculate the distance travelled.

- (a)** 12 m/s for 30 s. [1]

(b) 60 km/h for 2 hours.

[1]

4. Calculate the time taken.

(a) 400 m at 8 m/s.

[1]

(b) 150 km at 50 km/h.

[1]

SECTION 2 – UNITS

5. Convert each speed.

(a) 20 m/s into km/h

[1]

(b) 90 km/h into m/s

[1]

(c) 5 m/s into km/h

[1]

6. Convert each time.

(a) 30 minutes into hours

[1]

(b) 2 hours 15 minutes into hours

[1]

(c) 3 minutes into seconds

[1]

- 7.** A walker's journey is recorded below.

[3]

Time (s):	0	10	20	30
Distance (m):	0	15	30	45

Calculate the walker's speed, and state how you can tell it is constant.

- 8.** State what a horizontal line on a distance-time graph tells you about the object, and explain why.

[2]

- 9.** A cyclist travels 6 km in 15 minutes. Calculate her speed in km/h.

[3]

- 10.** A student calculates that his walking speed is 50 m/s. Explain why this answer must be wrong.

[2]

- 11.** A car travels at 15 m/s for 40 s. Calculate how far it goes, giving your answer in metres and in kilometres.

[2]

12. State the difference between average speed and instantaneous speed.

[2]

End of paper. Check every answer has a unit or a form the question actually asked for.